AN OBJECT WITH VANISHING COHOMOLOGY OF ITS COHOMOLOGY SHEAVES OVER A COFINAL SYSTEM IS THE DERIVED LIMIT OF ITS TRUNCATIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction to the cohomology sheaves and the use of the vanishing criterion are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Truncation functors tau_{>= -n} and the derived limit of a system of objects. Same chapter together with the chapter Derived Categories, the truncations of an object of D(O) and the construction of R lim of an inverse system of complexes. Supplies the object R lim tau_{>= -n}E and the canonical map from E to it whose being an isomorphism is asserted.
- Cohomology of a complex of O-modules at an object of the site, computed over a cofinal system of coverings. Same chapter, the vanishing criterion for cohomology of sheaves over a cofinal system of coverings, which is the hypothesis (2) of the statement. Supplies the input that makes the successive quotients of the limit vanish.

Source reference. Original lemma 0H72: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5337–5337. Navigation pointer invoked at proof line 5337; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6L: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5279–5279. Navigation pointer invoked at proof

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line 5371; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 08J5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5374–5374. Navigation pointer invoked at proof line 5374; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $E \in D(\mathcal{O})$. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset. Assume*

1. *every object of $\mathcal{C}$ has a covering whose members are elements of $\mathcal{B}$, and*
2. *for every $V \in \mathcal{B}$ there exist a function $p(V, -) : \mathbf{Z} \to \mathbf{Z}$ and a cofinal system Cov_V of coverings of V such that*
$$H^p(V_i, H^{m-p}(E)) = 0$$
for all $\{V_i \to V\} \in Cov_V$ and all integers p, m satisfying $p > p(V, m)$.

Then the map $E \to R\lim \tau_{\geq -n} E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O})$.

Proof. Set $K_n = \tau_{\geq -n} E$ and $K = R\lim K_n$. The canonical map $E \to K$ comes from the canonical maps $E \to K_n = \tau_{\geq -n} E$. We have to show that $E \to K$ induces an isomorphism $H^m(E) \to H^m(K)$ of cohomology sheaves. In the rest of the proof we fix m. If $n \geq -m$, then the map $E \to \tau_{\geq -n} E = K_n$ induces an isomorphism $H^m(E) \to H^m(K_n)$. To finish the proof it suffices to show that for every $V \in \mathcal{B}$ there exists an integer $n(V) \geq -m$ such that the map $H^m(K)(V) \to H^m(K_{n(V)})(V)$ is injective. Namely, then the composition
$$H^m(E)(V) \to H^m(K)(V) \to H^m(K_{n(V)})(V)$$
is a bijection and the second arrow is injective, hence the first arrow is bijective. By property (1) this will imply $H^m(E) \to H^m(K)$ is an isomorphism. Set
$$n(V) = 1 + \max\{-m, p(V, m-1) - m, -1 + p(V, m) - m, -2 + p(V, m+1) - m\}.$$
so that in any case $n(V) \geq -m$. Claim: the maps
$$H^{m-1}(V_i, K_{n+1}) \to H^{m-1}(V_i, K_n) \quad \text{and} \quad H^m(V_i, K_{n+1}) \to H^m(V_i, K_n)$$
are isomorphisms for $n \geq n(V)$ and $\{V_i \to V\} \in \mathrm{Cov}_V$. The claim implies conditions (1) and (2) of Lemma 0D6L are satisfied and hence implies the desired injectivity. Recall (Derived Categories, Remark 08J5) that we have distinguished triangles
$$H^{-n-1}(E)[n+1] \to K_{n+1} \to K_n \to H^{-n-1}(E)[n+2]$$
Looking at the associated long exact cohomology sequence the claim follows if
$$H^{m+n}(V_i, H^{-n-1}(E)), \quad H^{m+n+1}(V_i, H^{-n-1}(E)), \quad H^{m+n+2}(V_i, H^{-n-1}(E))$$
are zero for $n \geq n(V)$ and $\{V_i \to V\} \in \mathrm{Cov}_V$. This follows from our choice of $n(V)$ and the assumption in the lemma. □